Beyazit Yalcinkaya

beyazit@berkeley.edu
people.eecs.berkeley.edu/~beyazit/
Cory Hall, Berkeley, CA 94720

Research Interests

Formal methods, multi-task reinforcement learning, multi-agent systems, simulation, testing, verification, autonomous driving, representation learning, human-computer interaction.

Education

2021 - Present	**Ph.D. in Computer Science** Advised by Sanjit A. Seshia *University of California, Berkeley, CA, USA*
2015 - 2020	**B.Sc. in Computer Engineering** Valedictorian of the department *Middle East Technical University, Ankara, Turkiye*

Research Experience

2021 - Present	**Graduate Student Researcher** Supervised by Sanjit A. Seshia *University of California, Berkeley, CA, USA*
Summer 2026	**Research Intern** Supervised by Marcell Vazquez-Chanlatte *Zoox, Foster City, CA, USA*
Summer 2023	**Applied Scientist Intern** Supervised by Ankush Desai *Amazon Web Services, East Palo Alto, CA, USA*
Summer 2022	**Applied Scientist Intern** Supervised by Ankush Desai *Amazon Web Services, Cupertino, CA, USA*
2017 - 2020	**Undergraduate Research Assistant** Supervised by Ebru Aydin Gol *Middle East Technical University, Ankara, Turkiye*
Summer 2019	**Research Intern** Supervised by George Candea *École Polytechnique Fédérale de Lausanne, Lausanne, Switzerland*
Summer 2018	**Research Intern** Supervised by Björn B. Brandenburg *Max Planck Institute for Software Systems, Kaiserslautern, Germany*

Teaching Experience

Spring 2026	**Graduate Student Instructor** CS 189/289A Introduction to Machine Learning *University of California, Berkeley, CA, USA*
Fall 2022	**Graduate Student Instructor** EE/CS 149/C249A Introduction to Embedded Systems *University of California, Berkeley, CA, USA*
Fall 2017	**Undergraduate Teaching Assistant** CENG 230 Introduction to C Programming *Middle East Technical University, Ankara, Turkiye*

Refereed Conference and Journal Papers

[1] Abhinav Pomalapally, Arya Raeesi, Kevin Kai-Chun Chang, *Beyazit Yalcinkaya*, and Sanjit A. Seshia, **Scenario-Based Compositional Statistical Model Checking for Safety Specifications**, *International Conference on Runtime Verification (RV)*, 2026.

[2] *Beyazit Yalcinkaya*, Marcell Vazquez-Chanlatte, Ameesh Shah, Hanna Krasowski, and Sanjit A. Seshia, **Automata-Conditioned Cooperative Multi-Agent Reinforcement Learning**, *International Conference on Machine Learning (ICML)*, 2026.

[3] *Beyazit Yalcinkaya*, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, **Learning Task-Conditioned Policies with Natural Language and Graph Embeddings of Formal Specifications**, *International Conference on Neuro-symbolic Systems (NeuS)*, 2026.

[4] *Beyazit Yalcinkaya*, Niklas Lauffer, Marcell Vazquez-Chanlatte, and Sanjit A. Seshia, **Provably Correct Automata Embeddings for Optimal Automata-Conditioned Reinforcement Learning**, *International Conference on Neuro-symbolic Systems (NeuS)*, 2025.

[5] *Beyazit Yalcinkaya**, Niklas Lauffer*, Marcell Vazquez-Chanlatte*, and Sanjit A. Seshia, **Compositional Automata Embeddings for Goal-Conditioned Reinforcement Learning**, *Conference on Neural Information Processing Systems (NeurIPS)*, 2024. *Equal contribution.

[6] *Beyazit Yalcinkaya*, Hazem Torfah, Daniel J. Fremont, and Sanjit A. Seshia, **Compositional Simulation-Based Analysis of AI-Based Autonomous Systems for Markovian Specifications**, *International Conference on Runtime Verification (RV)*, 2023.

[7] *Beyazit Yalcinkaya*, Hazem Torfah, Ankush Desai, and Sanjit A. Seshia, **ULGEN: A Runtime Assurance Framework for Programming Safe Cyber-Physical Systems**, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2023.

[8] Niklas Lauffer*, *Beyazit Yalcinkaya**, Marcell Vazquez-Chanlatte, Ameesh Shah, and Sanjit A. Seshia, **Learning deterministic finite automata decompositions from examples and demonstrations**, *Formal Methods in Computer-Aided Design (FMCAD)*, 2022. *Equal contribution.

[9] Mert Ergurtuna, *Beyazit Yalcinkaya*, and Ebru Aydin Gol, **An automated system repair framework with signal temporal logic**, *Acta Informatica*, 2022.

[10] *Beyazit Yalcinkaya* and Ebru Aydin Gol, **Clock reduction in timed automata while preserving design parameters**, *International Conference on Formal Methods in Software Engineering (FormaliSE)*, 2019.

[11] *Beyazit Yalcinkaya*, Mitra Nasri, and Björn B. Brandenburg, **An exact schedulability test for non-preemptive self-suspending real-time tasks**, *Design, Automation and Test in Europe Conference (DATE)*, 2019.